

FITBUDDY AI

2. Requirement Analysis Phase

PROJECT REPORT DOCUMENT

Project: FitBuddy AI — Fitness Plan Generator

Group 1

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Objective

To define the functional, non-functional, data, safety and operating requirements for FitBuddy AI.

Prepared for academic project submission

2. Requirement Analysis Phase

Functional Requirements

- FR1: Accept user ID, name, age, weight, goal, intensity and experience level.
- FR2: Validate and normalize submitted profile values.
- FR3: Generate a seven-day workout plan with overview, daily focus, warm-up, main workout, cool-down and notes.
- FR4: Generate one concise nutrition/recovery/wellness tip.
- FR5: Store user profiles and plan records in a database.
- FR6: Accept feedback and revise the current plan while retaining the original plan.
- FR7: Provide REST endpoints to create, retrieve and revise plans and check service health.
- FR8: Restrict the all-users dashboard with HTTP Basic authentication.

Non-Functional Requirements

- Usability: clear form labels and readable plan output.
- Reliability: validation, error handling and database persistence.
- Maintainability: modular Python application structure.
- Compatibility: modern browser and Python 3.11+ environment.
- Security: keep API keys in environment configuration; change default admin credentials before deployment.
- Testability: support mock-AI mode and automated tests.

Input Constraints

- Age: 13–100; weight: greater than 20 kg and less than 400 kg; user ID: 2–80 characters; goals: weight loss, muscle gain, general wellness, flexibility or endurance; intensity: low, medium or high; experience: beginner, intermediate or advanced. Feedback length is limited to 1,200 characters.

Data Requirements

- User data: unique user ID, name, age, weight, goal, intensity, experience and timestamps.
- Plan data: original plan, revised plan if any, wellness tip, feedback, current-plan flag and timestamps.

External Dependencies

- FastAPI, Jinja2, SQLAlchemy, Pydantic, SQLite (default), Google GenAI SDK, environment configuration and pytest/httpx for testing. Gemini API key is required for live AI generation; mock mode supports local testing.

Safety & Privacy Requirements

- Provide general wellness information only; do not diagnose or prescribe treatment. Avoid extreme exercise, restrictive dieting and unsafe challenges. Recommend qualified professional guidance for injury or health concerns. Collect only information needed for the project and protect credentials.

Acceptance Criteria

- A valid profile produces a seven-day plan and wellness tip; invalid values are rejected; feedback produces an updated seven-day plan; plan data can be retrieved; health endpoint reports service status; admin route requires

credentials; mock-mode tests pass.

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